

PACKAGE INSERT

POLARINE TABLET/SYRUP

Composition:	Polarine Tablet	Polarine Syrup
Dexchlorpheniramine Maleate	2mg/tab	2mg/5ml
Pseudoephedrine HCl	60mg/tab	30mg/5ml

Product Description:

Tablet: Oval, deep convex, white, plain tablets without markings.
Syrup: A clear, dark brown syrup with plum flavour.

Pharmacodynamics:

Dexchlorpheniramine Maleate is an anti-histamine with anti-cholinergic properties. It is capable of producing a slight to moderate sedative effect. It appears to compete with histamine for receptor sites on effector cells and are of value clinically in the prevention and relief of many allergic manifestations. It has been demonstrated that the predominant activity of the optically active isomers of chlorpheniramine is the dextro-isomer. The dextro-isomer is approximately two times more active than the racemic compound. Since dexchlorpheniramine is the dextro-isomer and active moiety of chlorpheniramine, its action and uses is similar to those of chlorpheniramine. Pseudoephedrine acts as an indirect sympathomimetic agent by stimulating sympathetic (adrenergic) nerve endings to release noradrenaline. Noradrenaline in turn, stimulates alpha- and beta-receptors throughout the body. The action of pseudoephedrine hydrochloride is apparently more specific for the blood vessels of the upper respiratory tract and less specific for the blood vessels of the systemic circulation. The vasoconstriction elicited at these sites results in the shrinkage of swollen tissues in the sinuses and nasal passages. Pseudoephedrine acts on alpha-adrenergic receptors in the nasal mucosa and releases norepinephrine, hence producing vasoconstriction of the dilated nasal arterioles resulting in shrinkage of swollen nasal mucous membranes, reduction of tissue hyperemia, oedema and nasal decongestion and an increase in nasal airway patency. Drainage of sinus secretions is increased and obstructed eustachian ostia may be opened. Relaxation of bronchial smooth muscle by stimulation of beta-adrenergic receptors may also occur. Pseudoephedrine has direct and indirect sympathomimetic activity and is an effective upper respiratory tract decongestant. Pseudoephedrine is substantially less potent than ephedrine in producing both tachycardia and elevation in systolic blood pressure, and considerably less potent in causing stimulation of the central nervous system.

Pharmacokinetics:

Peak blood levels were achieved at an average time of 3 hours after administration. The half-life of dexchlorpheniramine maleate ranged from 20 to 24 hours. The drug when given is found to be extensively metabolized. The drug and metabolites were primarily excreted in the urine, with 19% of the dose appearing in 24 hours and a total of 34% in 48 hours. Pseudoephedrine is a physiologically active stereoisomer of ephedrine that acts directly on alpha-adrenergic receptors and to a lesser degree, beta-adrenergic receptors. The alpha-adrenergic effects are believed to result from the reduced production of cyclic adenosine 3', 5' monophosphate (cyclic 3',5' AMP) by inhibition of the enzyme adenylyl cyclase, whereas beta-adrenergic effects appear to be caused by the stimulation of adenylyl cyclase activity. Following oral administration of 30 mg of pseudoephedrine hydrochloride effects are noted within 30 minutes with peak activity occurring at approximately 1 to 4 hour after administration. Pseudoephedrine is absorbed from the gastrointestinal tract. It is resistant to metabolism by monoamine oxidase and is largely excreted unchanged in the urine together with small amounts of its hepatic metabolite. It has half-life of several hours; elimination is enhanced and half-life accordingly shorter in acid urine.

Indication:

It is indicated in nasal and upper respiratory tract congestion, common cold, acute sinusitis, allergic rhinitis.

Recommended Dosage:

Tablet: Adults: One tablet 3 times daily.
Syrup: Children 6 to 12 years: Half teaspoonful 6 hourly 3 times daily.
Adults : One teaspoonful 6 hourly 3 times daily.

Route of Administration: Oral

Contraindications:

This formula is contraindicated in patients exhibiting hypersensitivity to any of the components. Antihistamines are contraindicated in patients receiving monoamine oxidase inhibitors since these agents prolong and intensify the anticholinergic effects of anti-histamines. Antihistamines should not be used to treat lower respiratory tract symptoms. Sympathomimetic preparations are contraindicated in patients with severe hypertension, severe coronary artery disease and patients receiving MAO inhibitor therapy due to potentiation of the pressor effects of pseudoephedrine.

Warnings and Precautions:

Sympathomimetics should be used with caution in patients with hypertension, hyperthyroidism, diabetes mellitus and cardiovascular disease. Antihistamines should be used with caution in patients with narrow angle glaucoma, stenosing peptic ulcer, pyloroduodenal obstruction, symptomatic prostatic hypertrophy and bladder neck obstruction. Patients should be warned about engaging in activities requiring mental alertness such as driving a car or operating machinery. Not recommended for children below 2 years. Use with caution and on doctor's/pharmacist's advice in children 2 to 6 years of age.

Interactions with Other Medicaments:

Anti-histamines have additive effects with alcohol and other CNS depressants (hypnotics, sedatives, tranquilizers). Monoamine oxidase inhibitors prolong and intensify the anticholinergic effects of antihistamines and potentiate the pressor effects of sympathomimetics. Sympathomimetics may reduce the antihypertensive effects of methyldopa, mecamlamine, reserpine and veratrum alkaloids.

Pregnancy and Lactation:

Safety during pregnancy has not been established. It is not known whether Polarine Syrup is excreted in human milk and therefore caution should be exercised when administered to nursing mothers.

Use in Children:

Safety and effectiveness of Polarine Syrup have not been established in children below 2 years.

Use in Elderly:

Antihistamines may cause dizziness, sedation and hypotension in patients above 60 years. These patients are also more likely to have adverse reactions to sympathomimetics.

Side Effects:

General: Urticaria, drug rash, anaphylactic shock, photosensitivity, excessive perspiration, chills, dryness of mouth, nose and throat.
Cardiovascular system: Hypertension, headache, palpitation, tachycardia, extrasystoles.
Hematologic system: Hemolytic anemia, thrombocytopenia, agranulocytosis.
Nervous system: Sedation, sleepiness, dizziness, disturbed coordination, fatigue, confusion, restlessness, excitation, nervousness, tremor, irritability, insomnia, euphoria, paresthesia, blurred vision, diplopia, vertigo, tinnitus, acute labyrinthitis, hysteria, neuritis, convulsions, CNS depression, hallucinations.
Gastrointestinal system: Epigastric distress, anorexia, nausea, vomiting, diarrhoea, constipation.
Urinary system: Urinary frequency, difficult urination, urinary retention.
Respiratory system: Thickening of bronchial secretions, tightness of chest, wheezing, nasal stuffiness.

Symptoms and Treatment of Overdose:

Overdose reactions may vary from central nervous system depression to stimulation. Stimulation is particularly likely in children. Atropine-like signs and symptoms: dry mouth, fixed, dilated pupils, flushing, and gastrointestinal symptoms may also occur. If vomiting has not occurred spontaneously the conscious patient should be induced to vomit. This is best done by having the patient drink a glass of water or milk after which they should be made to gag. Precautions against aspiration must be taken, especially in infants and children. If vomiting is unsuccessful, gastric lavage is indicated within 3 hours after ingestion and even later if large amounts of milk or cream were given beforehand. Isotonic and half isotonic saline is the lavage solution of choice. Saline cathartics, such as milk of magnesia, by osmosis draw water into the bowel and therefore, are valuable for their action in rapid dilution of bowel content. Vasopressors may be used to treat hypotension.

Effects on Ability to Drive and Use Machine:

Patients should be warned about engaging in activities requiring mental alertness such as driving a car or operating machinery.

Storage Conditions: Store at or below 30°C. Protect from light.

Pack Sizes:

Tablet: Blister pack : 10 x 10, 50 x 10 and 100 x 10 tablets per strip.
Syrup : A bottle of 60ml, 100ml and 120ml.

Pack Sizes (export only):

Tablet : A bottle of 90 tablets and 1000 tablets.
Syrup : A bottle of 3.6 litres and 3.8 litres.

Shelf-life: 3 years.

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN/LOCAL GENERAL PRACTITIONER/PHARMACIST.

Manufacturer & Product Registration Holder:

Malaysia:
SUNWARD PHARMACEUTICAL SDN. BHD.
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